

UNIT 01

STOICHIOMETRY



STOICHIOMETRY

It is the branch of chemistry that deals with the relative quantities of reactants and products in chemical reactions. It involves calculations to determine the amounts of various substances involved.

EXPONENTIAL NOTATION

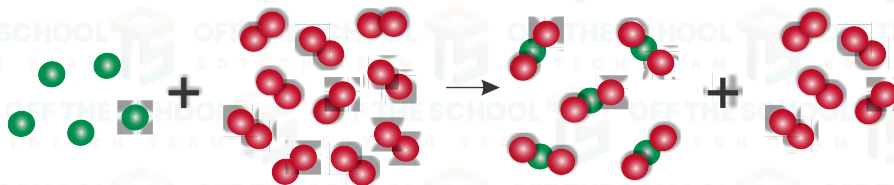
Also known as scientific notation, it's a way of expressing numbers that are too big or too small to be conveniently written in decimal form. It's composed of a coefficient and a power of 10.

MOLAR VOLUME

This is the volume occupied by one mole of any gas at a given temperature and pressure, commonly at standard temperature and pressure (STP). For ideal gases, this is about 22.4 liters per mole.

ATOM

It is the smallest unit of an element and enters into a chemical reaction. **OR** An atom is the smallest particle of an element that takes part in chemical change.

**ATOMIC NUMBER OR PROTON NUMBER (Z)**

The number of protons in the nucleus of an atom of an element is called as Atomic number.

It is equal to the number of electrons orbiting round the nucleus of that neutral atom. It is denoted by "Z".

MASS NUMBER OR NUCLEON NUMBER (A)

The sum of proton and neutron present in the nucleus of an atom is known as mass number.

$$\therefore A = P + n \quad \text{OR} \quad A = Z + n$$

ATOMIC MASS

The atomic mass of a single atom is simply its total mass and is typically expressed in atomic mass units or amu.

RELATIVE ATOMIC MASS (ATOMIC WEIGHT)

The relative atomic mass (A_r) of an element is defined as the ratio of the 'average mass of an atom' to 1/12th the mass of a $^{12}_6\text{C}$ atom.

SUBSTANCE

The purest form of matter is known as substance. They contain only one type of atom or molecule.

ELEMENT

An element is a substance which cannot be split into two or more simpler substances by ordinary chemical means.

OR

It is a pure substance or least complex form of matter and by simple chemical reaction cannot be converted into other simpler substance.



What is a mole in chemistry?



What is a mole?

One mole is the amount of substance that contains exactly $6.02214076 \times 10^{23}$ atoms, molecules or ions. This number is also known as 'Avogadro's number'. It's named after Italian scientist Amedeo Avogadro (left), a suggestion put forward by French scientist Jean Perrin to recognise Avogadro's work. 'Mole' derives from molecule – it's not related to the animal.



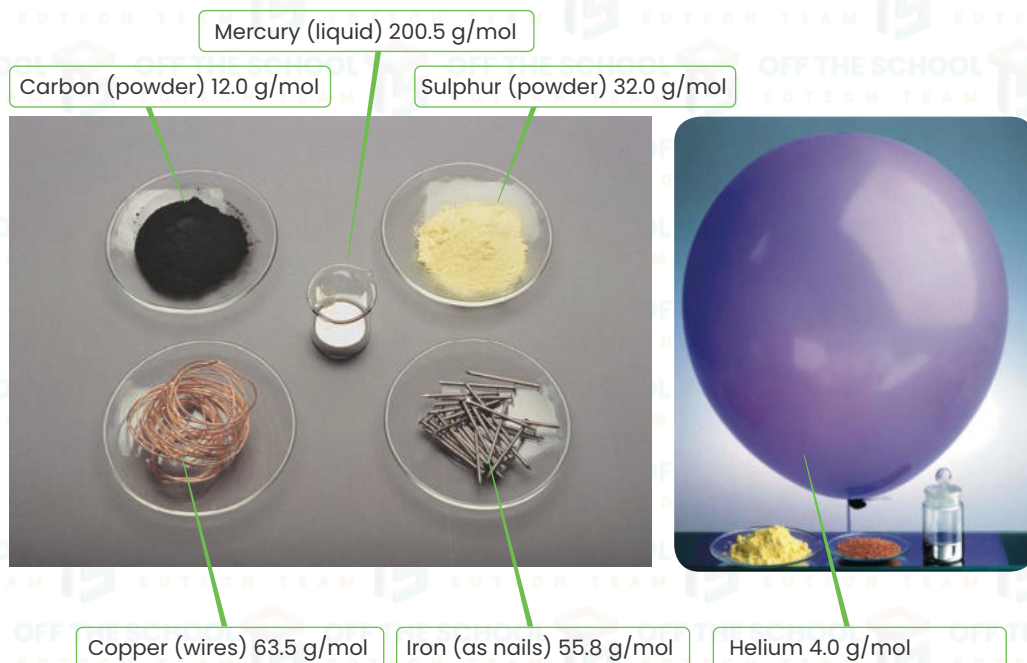
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The number of atoms, molecules or ions in one mole of a substance

Atoms, molecules and ions are very small and impossible for chemists to count. Using the mole makes it easier to talk about amounts of substances involved in reactions, by relating the mass of a substance to its atomic or molecular mass.

Amount of substance = mass (g) ÷ mass of 1 mole (g mol⁻¹)

One mole each of several common elements.



One mole contains a different mass for different substances

This makes sense if you think about it. Different substances will have atoms, molecules or ions which have different masses. Gold atoms have a greater mass than iron atoms, so the mass contained in one mole of gold atoms is greater.

03



Coins and moles: A useful analogy

Using moles to express amount of substance is analogous to weighing coin rolls to estimate the number of coins. In this analogy, the value of the coins is like mass (different for different coins), the number of coins is like number of atoms, and the rolls of coins are like moles of atoms.



MOLECULAR MASS

"The sum of all atomic masses presents in a molecule."

FORMULA MASS

"The sum of all atomic masses presents in a formula unit of ion."

AVOGADRO'S NUMBER

The Avogadro number (or Avogadro constant) is defined as the number of atoms in 12 grams of the C-12 isotope. **OR** One mole of a substance contains 6.02×10^{23} number of particles this number is called Avogadro number.



Avogadro's number:

602,214,100,000,000,000,000,000



Distance from Earth to sun (centimeters)

Average college tuition (U.S. dollars)

**THEORETICAL YIELD, ACTUAL YIELD, AND PERCENT YIELD****THEORETICAL YIELD**

In any chemical reaction, the theoretical yield is the maximum amount of product that can be generated as predicted by the chemical equation, assuming that every molecule of reactant converts into the product. It is based purely on stoichiometry, without taking into account the practical limitations. For example, if we react 2 moles of hydrogen with 1 mole of oxygen, stoichiometry tells us we can make 2 moles of water, because the balanced equation is;

**ACTUAL YIELD**

The actual yield is what is actually obtained from the reaction. It's often measured after the reaction has been carried out and the products have been isolated. This value is almost invariably lower than the theoretical yield due to factors such as incomplete reactions, by-products, side reactions, and losses during purification.

PERCENT YIELD

This is a measure of the efficiency of a reaction, calculated by dividing the actual yield by the theoretical yield and then multiplying by 100. It provides an insight into the losses that might occur in a practical setting and is vital for process optimization in industries.



REASON

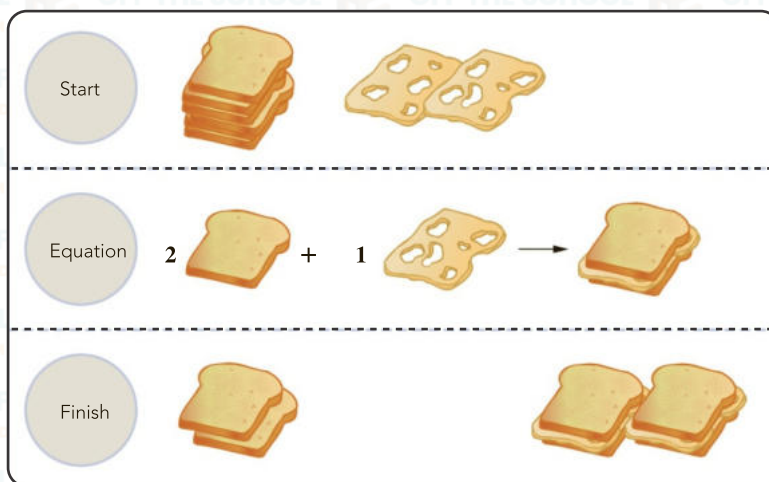
Why practical yield is less than theoretical yield?

Practical yield is often less than theoretical yield for various reasons. Some reactions are reversible, leading to an equilibrium where some reactants never convert into products. There may also be side reactions producing different products, and practical considerations like the purity of reactants, temperature control, measurement precision, and loss of product during extraction or transfer all play a part.

Limiting reactant:

The reactant which is present in least amount or entirely consumed first in a chemical reaction.

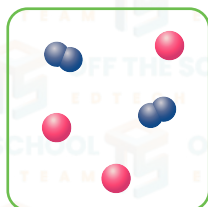
► **Limiting reactant analogy using cheese sandwiches** Start with six slices of bread, two slices of cheese, and the sandwich-making equation. Even though you have extra bread, you are limited to making two sandwiches by the amount of cheese you have on hand. Cheese is the limiting reactant.

**CONCEPT CHECK**

You perform the hypothetical reaction of an element, $X_2(g)$, with another element, $Y(g)$, to produce $XY(g)$.

- Write the balanced chemical equation for the reaction.
- If X_2 and Y were mixed in the quantities shown in the container on the left below and allowed to react, which of the three options is the correct representation of the contents of the container after the reaction has occurred?

Before reaction:



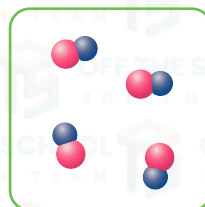
● = Atom X
● = Atom Y

After reaction:



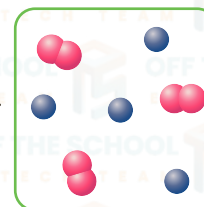
Option #1

or



Option #2

or



Option #3

- Using the information presented in Part b, identify the limiting reactant.

